1. (a)

AB\CD	00	01	11	10
00	0	1	0	1
01	0	x	1	0
11	1	x	1	0
10	0	x	x	x

质蕴含项：AD, BD, $C'D$, ABC', $AB'C$, $B'CD'$

实质蕴含项：BD, $C'D$, $B'CD'$, ABC'

$f = BD + C'D + B'CD' + ABC'$

(b)

AB\CD	00	01	11	10
00	1	1	0	1
01	0	1	x	x
11	1	0	0	x
10	1	1	1	0

质蕴含项：$B'+C'$, $A+B'+D$, $A+C'+D'$, $A'+B'+D'$, $A'+C'+D$

实质蕴含项：$A+B'+D$, $A+C'+D'$, $A'+B'+D'$, $A'+C'+D$

$f = (A+B'+D)(A+C'+D')(A'+B'+D')(A'+C'+D)$

2、

列表1

最小项	ABCD	
0	0000	✓
1	0001	✓
4	0100	✓
8	1000	✓
9	1001	✓
12	1100	✓
7	0111	✓
11	1011	✓
15	1111	✓

列表2

最小项	ABCD	
0, 1	000-	✓
0, 4	0-00	✓
0, 8	-000	✓
1, 9	-001	✓
4, 12	-100	✓
8, 9	100-	✓
8, 12	1-00	✓
9, 11	10-1	PI_3
7, 15	-111	PI_4
11, 15	1-11	PI_5

列表3

最小项	ABCD	
0,1,8,9	-00-	PI_1
0,4,8,12	--00	PI_2

	0	1	4	7	9	12
** PI_1	x	x			x	
** PI_2	x		x			x
PI_3					x	
** PI_4				x		
PI_5						

{PI_1, PI_2, PI_4}覆盖所有最小项，结束

$f = PI_1 + PI_2 + PI_4$

$= B'C' + C'D' + BCD$

3、(1) 划分法

$P1 = (AFH)(BCDEG)$

$P2 = (AH)(F)(BCDEG)$

$P3 = (AH)(F)(BD)(CEG)$

$P4 = P3$

(2) 蕴含表法

B	×						
C	×	~~FH~~					
D	×	CE	~~BE, HF~~				
E	×	~~CD, AF~~	BD, AH	~~HF~~			
F	~~AB, EF~~	×	×	×	×		
G	×	~~CD, FH~~	BD	~~DE, FH~~	AH	×	
H	EG	×	×	×	×	~~BH, FG~~	×
	A	B	C	D	E	F	G

$S_0 = \{A, H\}$ $S_1 = \{B, D\}$ $S_2 = \{C, E, G\}$ $S_3 = \{F\}$

化简后

现态	次态 $x=0$	次态 $x=1$	输出 $x=0$	输出 $x=1$
S_0	S_0	S_2	1	0
S_1	S_2	S_3	0	0
S_2	S_1	S_0	0	0
S_3	S_1	S_3	1	0

4. 规则1：S_1 adj S_3，S_2 adj S_3

规则2：S_0 adj S_2，S_2 adj S_3，S_0 adj S_1，S_1 adj S_3.